

Communications and Networks (Spring 2022)

Problem Set 6

Issued: Mar. 23 Due: Jun. 5

Readings:

Forouzan - Data Communications and Networking (5th ed) Chapter 8, 12, 6.1.1, 6.1.3

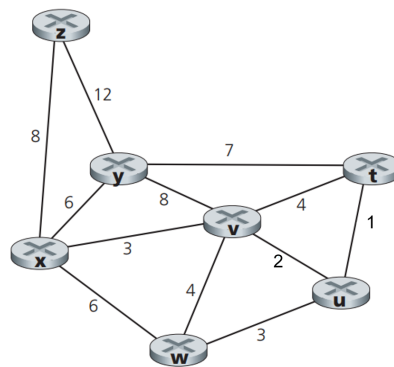
Sklar - Digital Communications Fundamental and Applications (2nd ed) 11.1.4, 11.3.1-11.3.2

Proakis - Digital Communications (5th ed) 16.1-16.2

Bertsekas - Data Network (2nd ed) 4.2.2

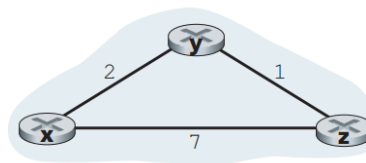
1. Consider an FDMA system with $K = 2$ users, in an AWGN channel, where user 1 is assigned a bandwidth $W_1 = \alpha W$ and user 2 is assigned a bandwidth $W_2 = (1 - \alpha)W$, where $0 \leq \alpha \leq 1$. Let P_1 and P_2 be the average powers of the two users, while $N_0/2$ is the double-sided power spectrum density of the additive noise.
 - a) Determine the capacities C_1 and C_2 of the two users and the total capacity $C = C_1 + C_2$ as a function of α .
 - b) Show that the maximum total capacity is achieved when $\alpha/(1 - \alpha) = P_1/P_2$.
2. (optional) In a pure ALOHA system, the channel bit rate is 2400 bits/s. Suppose that each terminal transmits a 100-bit message every **minute**.
 - a) Determine the maximum number of terminals that can use the channel without collision.
 - b) Repeat a) if slotted ALOHA is used.
3. In a multi-user slotted ALOHA system, the channel bit rate is 100 Mbps (1 M= 10^6). Assume that the arrival of packets follows a Poisson process. The size of a packet is 125 KB (1 Byte = 8 bits)
 - a) Determine the total traffic arrival rate (in packet/s) that ensures the system achieving the maximum throughput.
 - b) Determine the total traffic arrival rate (in packet/s) with which the system's throughput is equal to half of the maximum throughput.
4. In a pure ALOHA network with $G = 1/2$, how is the normalized throughput affected in each of the following cases? (Increasing or decreasing?)

- a) G is increased to 1.
- b) G is decreased to $1/4$.
- c) Repeat a) and b) if slotted ALOHA with $G = 1/2$ is used.
5. List the strategies in CSMA/CA that are used to avoid collision. There is no acknowledgment mechanism in CSMA/CD, but we need this mechanism in CSMA/CA. Explain the reason briefly.
6. Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from x to all network nodes. Show how the algorithm works by computing the following table.



Step	start S	D(z), p(z)	D(y), p(y)	D(v), p(v)	D(w), p(w)	D(u), p(u)	D(t), p(t)
0	x						

7. Consider the network shown below, and assume that each node initially knows the costs to each of its neighbors. Consider the distance-vector algorithm and show the distance table at node x (Hint: the initial distance table at node x is given below)



		cost to		
from		x	y	z
	x	0	2	7
	y	∞	∞	∞
	z	∞	∞	∞

Time →